

Multi-compartment models

and the cable equation

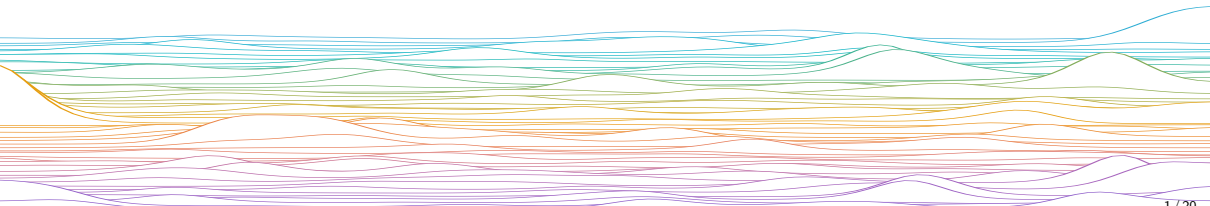
Computational Neuroscience

University of Bristol

M Rule

Learning outcomes:

- ▶ Describe ***multi-compartmental*** models and how these relate to the cable equation
- ▶ Discuss what the cable equation is and what features of neuronal electronic dynamics it captures



Point neuron

- ▶ One compartment for the whole cell, axons, dendrites, etc.

Two compartment

- ▶ One compartment for dendrites, one compartment for soma+axons

Multi-compartment models

- ▶ ...

Passive membrane

- ▶ with membrane resistance and capacitance R_m and C_m
- ▶ rest voltage V_r ,
- ▶ and external input current I :

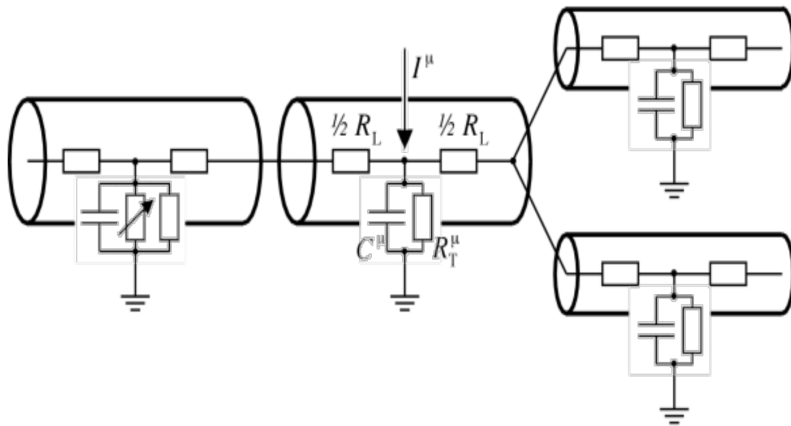
$$C_m \dot{v} = \frac{1}{R_m} (V_r - v)$$

Electrical connection conductance g between two compartments

- ▶ with voltages v_1 and v_2 ,
- ▶ capacitances C_1 and C_2 ,
- ▶ and axial conductance g_α :

$$C_1 \dot{v}_1 = g_\alpha (v_2 - v_1)$$

$$C_2 \dot{v}_2 = g_\alpha (v_1 - v_2)$$



Adapted by Ani Boja, 2025

Electrical connections aren't that different from ionic conductances

Instead of being pulled toward an ionic reversal potential, voltage is pulled toward the voltages in another connected compartment

Instead of ionic conductance, we have an axial conductance

Three compartments

$$C_m \dot{v}_1 = g(v_2 - v_1) + I_1$$

$$C_m \dot{v}_2 = g(v_1 - v_2) + g(v_3 - v_2) + I_2$$

$$= g(v_1 + v_3 - 2v_2) + I_2$$

$$C_m \dot{v}_3 = g(v_2 - v_3) + I_3$$

Electrical graphs

Implementing multi-compartment models: Not much harder than simulating a network of point neurons

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Implementing multi-compartment models: Not much harder than simulating a network of point neurons

Each compartment acts like its own point neuron, e.g. the i^{th} compartment:

$$C_i \dot{v}_i = g_i^\ell (E_i^\ell - v_i) + I_i^{\text{membrane}} \quad \text{currents through membrane}$$
$$+ \sum_{j \neq i} g_{ij}^{\text{axial}} (v_j - v_i) \quad \text{currents to/from connected compartments}$$

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Currents from connected compartments couple (gap-junction synapses similar)

Electrical graphs

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$$\begin{aligned} C_i \dot{v}_i &= g_i^\ell (E_i^\ell - v_i) + I_i^{\text{membrane}} && \text{currents through membrane} \\ &+ \sum_{j \neq i} g_{ij}^{\text{axial}} (v_j - v_i) && \text{currents to/from connected compartments} \end{aligned}$$

Currents from connected compartments couple (gap-junction synapses similar)

Vector form, using $\circ$ for element-wise multiplication

$$\begin{aligned} C \circ \dot{\mathbf{v}} &= \mathbf{g}_\ell \circ (\mathbf{E}_\ell - \mathbf{v}) + \mathbf{I}_{\text{membrane}}(\dots) + \mathbf{G}_{\text{axial}} \mathbf{v} \\ \{\mathbf{G}_{\text{axial}}\}_{ij} : &\begin{cases} 0 & i \neq j \text{ disconnected compartments} \\ g_{ij} = g_{ji} \geq 0 & i \neq j \text{ connected compartments} \\ -\sum_{k \neq i} g_{ik} & i = j \text{ (diagonal)} \end{cases} \end{aligned}$$

Three compartments

$$C_m \dot{v}_1 = g(v_2 - v_1) + I_1$$

$$\begin{aligned} C_m \dot{v}_2 &= g(v_1 - v_2) + g(v_3 - v_2) + I_2 \\ &= g(v_1 + v_3 - 2v_2) + I_2 \end{aligned}$$

$$C_m \dot{v}_3 = g(v_2 - v_3) + I_3$$

∞ compartments?

Cable Equation

Cable Equation: Spatially continuous limit of multi-compartment model

Consider a small length Δ_x

c_m, g_ℓ, r_α : “specific” (per unit length)

- ▶ Membrane capacitance $c_m \cdot \Delta_x$
- ▶ Membrane leak conductance $g_\ell \cdot \Delta_x$
- ▶ Axial resistance $r_\alpha \cdot \Delta_x$

$$c_m \Delta_x \dot{v}(x) = \frac{1}{r_\alpha \Delta_x} [v(x - \Delta_x) + v(x + \Delta_x) - 2v(x)] + g_\ell \Delta_x (E_\ell - v) + \dots$$

$$c_m \dot{v}(x) = \frac{1}{r_\alpha} \frac{v(x - \Delta_x) + v(x + \Delta_x) - 2v(x)}{\Delta_x^2} + g_\ell (E_\ell - v) + \dots$$

$$c_m \dot{v}(x) = \frac{1}{r_\alpha} \frac{\partial^2}{\partial x^2} v(x) + g_\ell (E_\ell - v) + \dots$$

For those who prefer terse, slightly ambiguous notation

Consider a small length Δ

c, g, r : “specific” (per unit length)

- ▶ Membrane capacitance $c \cdot \Delta$
- ▶ Membrane leak conductance $g_\ell \cdot \Delta$
- ▶ Axial resistance $r \cdot \Delta$

$$c \Delta \dot{v} = \frac{1}{r \Delta} (v_{-\Delta} + v_{+\Delta} - 2v) + g_\ell \Delta (E_\ell - v) + \dots$$

$$c \dot{v} = \frac{1}{r} \left(\frac{v_{-\Delta} + v_{+\Delta} - 2v}{\Delta^2} \right) + g_\ell (E_\ell - v) + \dots$$

$$c \dot{v} = \frac{1}{r} v_{xx} + g_\ell (E_\ell - v) + \dots$$

where $v_{xx} := \partial_x^2 v(x, t)$ is partial differential equation notation for the second spatial derivative.

Compare to a heat (diffusion) equation.

Multi-compartment models

- ▶ We usually don't simulate the whole cable equation
 - ▶ We break the neuron down into larger compartments
-
- ▶ The mathematics of the cable equation is not (currently) on the 2025 draft exam or coursework.
 - ▶ But, synthesising insights from multi-compartment models is.
 - ▶ (All statements about assessments are provisional, things may change)



show them a simulated model

Modelling Scales

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin–Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch–Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
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Tractability



Phenomenological,
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